

IBVAP — Streamlit Dashboard & UI Specifications

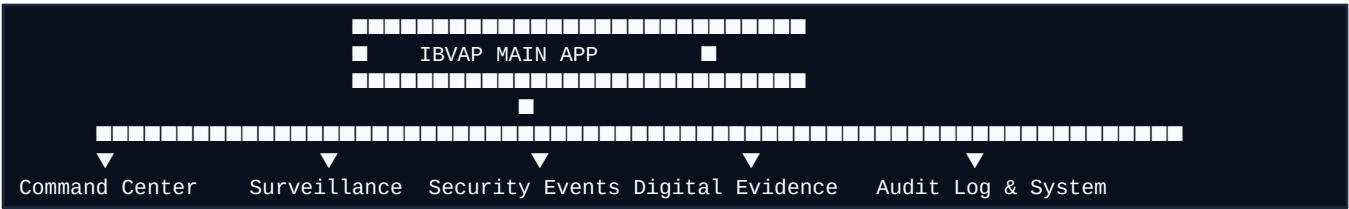
10. Dashboard Layout, Navigation & Page Breakdown

1. Web Application Architecture

The IBVAP dashboard is built using Streamlit with custom CSS glassmorphism, responsive grid layouts, SVG icons, and Plotly interactive charts.

- **Theme Configuration:** Custom Dark Navy Theme defined in `streamlit_app/.streamlit/config.toml` (Primary background `#060b14`, Card background `#0e1726`, Accent cyan `#00f5d4`).
- **Typography:** **Plus Jakarta Sans** Google Font loaded via CSS `@import`.
- **Navigation Options:**
 - **Single-File App:** `src/app.py` (Full feature implementation in 1605 lines).
 - **Modularized App:** `streamlit_app/app.py` + `streamlit_app/pages/` modules.

2. Detailed Page-by-Page Feature Matrix



Page 1: Command Center

- **Purpose:** Executive high-level operational surveillance dashboard.
- **Components:**
 - **4 Top KPI Cards:** Total Intrusion Events (3), Unique Person Tracks (3), Threat Level (HIGH), Digital Evidence Items Captured (3).
 - **Threat Status Card:** High-risk breach alert card with glowing red border and threat description.
 - **System Health Grid:** Real-time online indicators for AI Engine, Tracking Engine, Zone Monitor, Event Logger, Evidence Capture, and Audit System.
 - **AI Security Pipeline Component:** Step-by-step visual vector flowchart mapping processing from raw camera feed to audit logging.
 - **Interactive Plotly Charts:** Risk Level Bar Chart (HIGH/MEDIUM/LOW distribution) + Event Frame Scatter Plot.
 - **Recent Events Table:** Preview list of latest breach entries.

Page 2: Surveillance

- **Purpose:** Video monitoring, zone visualizer, and event timeline inspector.
- **Components:**

- **Processed Video Stream Player:** Embeds and plays [output/final_intrusion_video.mp4](#).
- **Restricted Zone Visualizer:** Interactive SVG canvas overlay rendering the polygon restricted border area and active target markers (e.g. [T:2](#), [T:7](#), [T:36](#)).
- **Interactive Event Timeline:** Button list of timestamped events. Clicking an event selects it for deep inspection.
- **Selected Event Inspector:** Detailed sidebar card listing Event ID, Track ID, Frame, Timestamp, and direct link to evidence.

Page 3: Security Events

- **Purpose:** Master data management table for security incidents.
- **Components:**
 - **Filter Controls:** Free text search box (matches Event ID, Track, Timestamp), Risk Dropdown ([ALL](#), [HIGH](#), [MEDIUM](#), [LOW](#)), Status Dropdown ([Confirmed](#), [Pending](#), [Resolved](#)), and Sorting Dropdown ([Frame](#), [Timestamp](#), [Risk](#)).
 - **Data Table:** Custom styled table rendering all recorded security breach rows.
 - **Event Summary Modal:** Interactive pop-up card displaying complete event metadata, evidence status, and JSON download capability.

Page 4: Digital Evidence

- **Purpose:** Visual evidence gallery and snapshot inspection.
- **Components:**
 - **Evidence Grid:** Responsive 2-column image card layout displaying snapshot images alongside track metadata.
 - **Action Toolbar:** Zoom button, direct JPEG image download button, and jump-to-event button.
 - **High-Resolution Modal Viewer:** Full-width image expansion for forensic analysis.

Page 5: Audit Log

- **Purpose:** Cryptographic audit trail and integrity verification hub.
- **Components:**
 - **Structured Log Table:** Complete history table with exact timestamps, frame indices, and creation metadata.
 - **One-Click CSV Export:** Downloads [ibvap_security_audit_log.csv](#).
 - **Live Cryptographic Verification Panel:** Executes [verify_secure_audit_log\(\)](#) and displays root hash, verification status, and chain validation.
 - **Blockchain Ledger Status:** Displays [blockchain_ledger.json](#) block count and SHA-256 block-chain status.
 - **Cybersecurity Architecture Comparison:** Deployed CSV/SHA-256 architecture vs Future Enterprise Blockchain Roadmap.

Page 6: System

- **Purpose:** System architecture specs, module inventory, and environment health.
- **Components:**
 - **Module Status Cards:** Specifications for YOLO11, ByteTrack, OpenCV, Polygon Zone, Risk Engine, CSV Logger, and Streamlit.
 - **End-to-End Pipeline View:** Complete processing flowchart.
 - **Architecture Specs:** Technical comparison of core pipeline vs enterprise multi-camera deployment roadmap.